Chapter 8 Motion and energy

EXTRA WORKSHEET 1: Journey planning calculations

In planning a trip, it helps to consider the distances to be travelled when estimating travel time. Table 1 below shows the distance between various destinations in Australia, assuming the most direct route is taken. Use this data to answer the questions below. Round your answers correct to one decimal place.

Table 1

Travel Distance (km)	Adelaide	Brisbane	Canberra	Darwin	Melbourne	Perth	Sydney
Adelaide	-	2062	1205	3025	725	2707	1425
Albury	913	1545	345	3937	310	3620	563
Alice Springs	1534	2945	2680	1490	2259	3935	2958
Brisbane	2062	-	1269	3400	1685	4363	982
Broken Hill	509	1553	1072	3127	820	2810	1175
Broome	4270	4645	4975	1875	4995	2258	5110
Cairns	3382	1697	2920	2885	3008	6050	2678
Canberra	1200	1268	-	3915	655	3910	286
Darwin	3024	3400	4170	-	3750	4162	3994
Kalgoorlie	2188	3635	3392	4760	2913	598	3465
Mackay	2665	980	2205	2913	2290	5275	1940
Melbourne	724	1685	653	3749	-	3431	873
Mt Gambier	454	2071	1069	3479	482	3161	1289
Newcastle	1600	807	461	3820	1049	4159	175
Perth	2707	4363	3910	4163	3432	-	3985
Sydney	1424	981	286	3994	873	4131	-
Townsville	3038	1351	2577	2544	2662	5911	2313
Wollongong	1344	1022	236	4076	793	3392	81

Questions

1 Kelly catches a train from Sydney to Woolongong. The trip takes her 1hr 45 mins. Calculate the average speed of the train in km/h.

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2 Tien is a keen cyclist. He cycles from his home in Adelaide to his friend's house in Mt Gambier. Given that his average cycling speed is 28 km/h, **calculate** his expected travel time in hours.

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3 Sebastian travels on an overnight bus from Sydney to visit his brother in Melbourne. Given that the average speed of the bus is 65 km/h, **calculate** the number of hours Sebastian can expect to travel on the bus**.**

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4 Joseph travels from Broken Hill to Brisbane, picks up his friend, Sally and travels from Brisbane to Cairns for a beach holiday.

a Given that the trip took two and a half days, **calculate** the average speed of Joseph and Sally for the journey (in km/h).

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b Do you think that Joseph and Sally would have travelled at the exact speed calculated in your answer above for much of the journey? **Explain** your response**.**

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5 Nieve is driving a truck from Darwin to Alice Springs. She accelerates at a constant rate for the first 3 minutes of her motion to reach a speed of 90 km/h.

a **Calculate** Nieve's speed in m/s.

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b **Calculate** the average acceleration of the truck in these first 3 minutes (in m/s/s).

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c **Calculate** the distance that Nieve travels as the truck accelerates. Assume her average speed, in m/s, during acceleration is her top speed ÷ 2.

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d Assuming that Nieve is able to travel the remainder of the distance to Darwin at 90 km/h, **calculate** how much longer the trip will take.

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